

Yuheng Li

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Education

University of California, San Diego

Master of Science in Computer Science

La Jolla, CA

Expected Dec. 2026

University of California, Los Angeles

Bachelor of Science in Mathematics of Computation

Los Angeles, CA

Jun. 2025

Skills

Programming Languages: Python, C/C++, Java, SQL, MATLAB, Bash

Frameworks & Tools: PyTorch, Hugging Face, LangChain, OpenAI Agents SDK, Git, FastAPI, AWS

AI/ML: NLP, LLMs, RAG, AI Agents, Post-Training, Machine Learning Systems

Experience

Molex — Machine Learning Engineer Intern

Fremont, CA — Jun. 2026 – Sep. 2026

- Developed a domain-specific **AI Agent** with **LangChain**, **MCP**, and **RAG** to translate natural-language requests into multi-step Zemax workflows grounded in live simulation state and **1,300+** ZOS-API references.
- Engineered closed-loop LLM tool orchestration for live Zemax state inspection, code execution, and result verification, using trace-based **failure attribution** to improve task success rate by approximately **40%**.
- Designed a simulator-grounded **evaluation framework** with **100+** tasks, using reference-code executions as ground truth to evaluate a DeepSeek V4 Flash-powered agent, which achieved over **90%** end-to-end task accuracy.
- Integrated an agent backend with a CopilotKit frontend via AG-UI, enabling real-time execution visibility, session management, and **context engineering** through conversation history compression.

Advance.AI — Machine Learning Engineer Intern

Singapore — Jun. 2025 – Sep. 2025

- Designed an **LLM-driven** data synthesis pipeline using prompt engineering to convert OCR outputs into structured training data for ID documents, scaling labeled samples to **50K+** while reducing annotation time by **~95%**.
- Developed **layout-agnostic** post-processing quality filters to validate LLM outputs and curate the synthetic dataset, achieving **~95%** label accuracy across **~50%** more document layouts with no additional verification effort.
- Fine-tuned a **multimodal** LayoutLM in **PyTorch** on the synthetic dataset by fusing 2D positional embeddings of each token's spatial coordinates with textual signals, increasing field-level **F1-score** by **11%**.

Goldstate Securities Co., Ltd. — Data Scientist Intern

Shenzhen, China — Jul. 2024 – Sep. 2024

- Collaborated with traders to develop data processing pipelines and visualization tools covering approximately **1,800** A-share stocks, enabling efficient exploration of market trends, factor signals, and trading opportunities.
- Built an **LLM-driven** portfolio analysis and daily strategy reporting system.

Projects

Schema Linking for Text-to-SQL | SFT, LoRA, BM25, Hugging Face

Apr. 2026 – Jun. 2026

- Designed a schema-linking pipeline combining LLM-generated **data augmentation** with **BM25** retrieval, dynamically selecting relevant tables under a fixed token budget and achieving **96%** column recall.
- Fine-tuned **Qwen3-1.7B** with **LoRA** on augmented schema-linking data, achieving **0.751** combined F1 (equal-weighted table/column) vs. **0.722** for **DeepSeek V4 Flash** (284B) on the same evaluation.

RAG over Technical Documentation | RAG, FAISS, Reranking, LangChain

Sep. 2025 – Dec. 2025

- Developed a **RAG** system that combines lexical and semantic retrieval with reranking, improving evidence retrieval quality and achieving approximately **0.87 Precision@5** and **0.81 Recall@5**.
- Built an **LLM-as-a-judge** evaluation pipeline to measure correctness, faithfulness, and completeness, helping identify retrieval failures separately from generation errors.

Research

SourceBench [arXiv]

Jan. 2026 – May. 2026

- “SourceBench: Can AI Answers Reference Quality Web Sources?” — Hexi Jin*, Stephen Liu*, **Yuheng Li***, Simran Malik, Yiying Zhang. **NeurIPS 2026**. (Accepted, Poster). *Co-first authors.

Decode-Free Information Extraction [GitHub]

Jan. 2026 – Mar. 2026

- Proposed a single-pass information extraction method that classifies passage relevance from **LLM hidden states** at EOS positions, bypassing autoregressive decoding; evaluated on **HotpotQA**.
- Applied **LoRA** fine-tuning on Qwen3-0.6B and Qwen3-Embedding-0.6B backbones, achieving **0.89 F1** with a **14%** relative improvement over the finetuned encoder baseline on multi-hop evidence selection.